

LEO NG

Electrical Engineering & Computer Science Student

+61 452 385 582 | leongheileong@gmail.com | linkedin.com/in/hei-leong-leo-ng | github.com/Slitre | slitre.github.io

PROFILE

- Bachelor of Engineering (Electrical) and Computer Science student at UNSW with a Distinction average, seeking engineering internships across electrical, electronics, embedded systems and automation roles.
- Hands-on experience designing and debugging embedded sensing systems, PCB hardware and hardware/software integrations, alongside programming and automation projects.
- Strong practical problem-solving, communication and teamwork skills developed through engineering projects and customer-facing operations work.

EDUCATION

Bachelor of Engineering (Electrical) and Computer Science

2023 – Estimated Aug 2027

UNSW Sydney — Distinction average

ENGINEERING PROJECTS

Embedded Sensing System / Piano Teaching Aid (LEMAT)

May – Aug 2026

UNSW Course Engineering Project

- Designed an RP2040-based sensing system using 13 VL53L1X time-of-flight sensors for real-time key and hand-position detection.
- Integrated multiple I²C sensors using XSHUT control and address assignment, debugging communication timeouts, startup conflicts and inconsistent readings.
- Designed PCB schematics and layouts in KiCad incorporating sensor interfaces, power regulation, connectors, LEDs and microcontroller circuitry.
- Tested and troubleshot power, signal and sensor behaviour, including battery-related LED instability, audio noise and reflective-surface sensing issues.

Technologies: RP2040, C/MicroPython, VL53L1X, I²C, KiCad, PCB Design, Embedded Systems

Custom Flip Phone – Hardware & Software Integration

Aug 2026 – Present

Personal Engineering Project

- Developing a compact flip-phone platform integrating an Orange Pi 3B, 4.97-inch MIPI DSI display, keypad controls and Android-based software.
- Investigated display-controller pinouts, MIPI DSI lane configuration, voltage requirements and Linux/Android panel-driver integration for hardware bring-up.
- Built a Kotlin/Jetpack Compose launcher with physical-key navigation, connecting software behaviour to custom hardware constraints.

Technologies: Orange Pi, MIPI DSI, Android, Kotlin, Linux, Hardware/Software Integration

Sankey Chart Dashboard App

Sep 2026 – Present

UNSW COMP3900 Capstone with Atlassian — Product Owner

- Working in a five-person team to develop a reusable React Sankey chart widget for Jira dashboards using Atlassian Forge.
- Coordinate client communication and help define requirements and project scope as the team's product owner.
- Contribute to design planning for data validation, readable flow visualisation, interactivity, accessibility and responsive dashboard layouts.

Technologies: React, JavaScript/TypeScript, Atlassian Forge, Jira, Git

Automation & AI Solutions Project – Gate Gourmet Japan

Aug – Sep 2024

UNSW CDEV3300 Industry-linked Project

- Analysed rostering and workforce-scheduling inefficiencies arising from manual Excel-based processes in a small project team.
- Developed a Python-based prototype to automate roster-processing workflows and reduce manual administrative effort.
- Evaluated automation and AI solutions and presented technical findings, recommendations and implementation considerations.

Technologies: Python, Excel, Automation Research

EMPLOYMENT

Restaurant Team Member & Operations Support

Dec 2022 – Present

Pho Sam Lane Cove

- Support day-to-day operations including stocktake, inventory tracking, supplier invoicing and workflow coordination in a fast-paced environment.
- Provide technical troubleshooting for POS systems, devices and other technology-related issues.

Freelance Website Developer

Apr – Nov 2024

Common Realty, Chatswood

- Developed and customised a client-facing website using Wix and Wix Velo, translating business requirements into website functionality.
- Implemented responsive pages, custom JavaScript functionality and backend integrations to support business and client workflows.

TECHNICAL SKILLS

Programming: C, Python, JavaScript, Shell, Kotlin

Engineering Tools: KiCad, MATLAB, Simulink, LTspice, Git, Linux

Web & Automation: React, Atlassian Forge, Wix Velo, CSS, automation scripting

Embedded & Hardware: RP2040, I²C, sensor integration, PCB design, power regulation, hardware/firmware troubleshooting

Testing & Integration: Circuit testing, debugging, component selection, hardware validation, hardware/software integration

Languages: English (Fluent), Cantonese (Native), Mandarin (Fluent), Japanese (Basic), German (A2)